Presentation — seven slides

Light theme; the NyaySetu mark top left. One idea per slide, large numbers, and nothing on a slide that cannot be shown on the live system. Speaker notes are under each slide.

Slide 1 · Title

NyaySetu — Proof, not paperwork Tamper-evident evidence, summons and bail records for Maharashtra's criminal courts. Governance and civic technology · SDG 16 · Live at nyaysetu-project.vercel.app Developed by Sai Randive

Visual: phone capturing evidence → fingerprint → ledger → courtroom.

> Notes: one sentence — "We make it possible for anyone, including the accused, to check that the evidence in court is the evidence that was collected."

Slide 2 · The problem

- More than **50 lakh** cases pending in Maharashtra's district courts, most of them criminal (NJDG)
- Since **1 July 2024**: BNSS §105 video, BNSS §176(3) forensics at the scene, BSA §63 hash certificate
- Three questions settled by someone's word: **Was the evidence altered? Was the summons served? Is bail being kept?**

Visual: a paper malkhana register beside a forwarded video file.

> Notes: a register records that an item moved, not that it is the same item.

Slide 2a · One real case

Pune, May 2024 — the swapped blood sample. In a crash that killed two people, the driver's sample was allegedly replaced inside a hospital. Caught only by a tip-off, a second sample and DNA, days later. **With NyaySetu:** the seal serial bound at collection; the substitute is refused at the laboratory the same day.

Visual: the two-lane replay from `/case-study`, stopped at the laboratory step.

Slide 3 · The solution

Capture → Anchor → Hand over → Verify → Produce in court

- SHA-256 fingerprint on the officer's phone, **before upload**; works offline
- Fingerprint on a public blockchain; **personal data never on chain**
- Every hand-over re-checks; a mismatch is **refused and recorded permanently**
- **Anyone** can verify a file — by reference, QR code, or the file alone
- BSA §63 certificate generated from the record, with a QR code

Visual: five screenshots in a row.

Slide 4 · What you will see (live)

1. Case replay: the Pune sample swap, caught at the first hand-over
2. Tamper lab: change one bit → the fingerprint changes completely
3. Public verification, no account: original ✅, one-pixel copy ❌
4. Laboratory receives a different seal or file → refused, anchored, alert raised
5. Accused hears obligations read aloud in Marathi
6. Judge sees the bail geo-fence and check-ins on a map
7. §63 certificate PDF → scan the QR → verification opens

Slide 5 · Architecture and quality

Phone / web (React, PWA, Expo offline queue) → API (Node, TypeScript, verification state machine, role and per-case access) → PostgreSQL with row-level security on all 48 tables · encrypted IPFS storage · three verified Solidity contracts on Ethereum · Python models labelled as decision support.

Measured	Result
One-byte tampering detected (200 files)	**100%**
Unauthorised access blocked (258 attempts)	**100%**
Items lost with 35% of chain writes failing	**0**
Automated tests	**183**, plus 44 pages checked for accessibility

Slide 6 · Feasibility and scale

- **Deployed today**; every subsystem's health is public
- **Cost measured:** 3.6 lakh gas to register an item, 1.56 lakh per hand-over — fixed, whatever the file size

- **State rollout:** a permissioned EVM ledger with nodes held by police, judiciary, forensic directorate and legal-services authority — no gas fees, no single party in control
- **Integration path:** CCTNS / ICJS / e-Courts through the API (designed); complements e-Sakshya by anchoring its recordings

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Slide 7 · Impact and next step

- Access to justice (SDG 16.3): defence and accused verify independently, in their language
- Accountable institutions (16.6): failures cannot be hidden
- Public information (16.10): live transparency page, no personal data

Pilot: Shivajinagar Police Station, Regional Forensic Laboratory Pune, one Sessions Court, one legal-aid clinic — three months.

Honest limit: a matching fingerprint proves a file is unchanged, not that it is true.